

# Task 21 – Basic SQL SELECT Queries

## Today's Task Explanation Report

### Objective

The objective of this task was to practice basic SQL operations using the Northwind dataset in MySQL Workbench. The task focused on retrieving, filtering, sorting, counting, and combining relational data.

### Tools and Dataset

| Item            | Details                                                           |
|-----------------|-------------------------------------------------------------------|
| Database Tool   | MySQL Workbench                                                   |
| Dataset         | Northwind                                                         |
| Tables Used     | customer, products, orders, order_details                         |
| Main SQL Topics | SELECT, WHERE, ORDER BY, COUNT(), JOIN, calculated columns, LIMIT |

### Work Performed

1. Imported and worked with the Northwind customer, product, order, and order detail tables.
2. Used SELECT to retrieve complete and selected records.
3. Used WHERE to filter customers and products based on conditions.
4. Used ORDER BY to sort customer information alphabetically.
5. Used JOIN operations to combine information from multiple related tables.
6. Created a calculated total amount using quantity × unit price.
7. Used COUNT() to determine the number of customer records.

### Important SQL Queries Practiced

```
SELECT * FROM customer;

SELECT customerID, companyName, city FROM customer;

SELECT customerID, companyName, city, country FROM customer WHERE country = 'Germany';

SELECT customerID, companyName, city FROM customer ORDER BY companyName ASC;

SELECT id, product_name, list_price FROM products WHERE list_price > 50;

SELECT o.id AS order_id, c.companyName, p.product_name,
od.quantity, od.unit_price, (od.quantity * od.unit_price) AS total_amount
FROM orders o
JOIN customer c ON o.customer_id = c.customerID
JOIN order_details od ON o.id = od.order_id
JOIN products p ON od.product_id = p.id
LIMIT 20;

SELECT COUNT(*) AS total_customers FROM customer;
```

### Key Learnings

- Understood how SELECT retrieves data from database tables.

- Learned how WHERE filters records according to conditions.
- Practiced sorting with ORDER BY.
- Learned how JOIN connects related tables.
- Created calculated fields inside a query.
- Used COUNT() for basic record counting.
- Improved practical experience with relational database tables in MySQL Workbench.

## **Conclusion**

Task 21 provided practical experience with fundamental SQL commands and relational data. Working with the Northwind dataset strengthened skills in data retrieval, filtering, sorting, aggregation, calculated fields, and table joins using MySQL Workbench.